GEOG 4/5/7 9073: Environmental Analysis in R

Week 04.01: Writing functions and reproducible code

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Today's schedule

- Open discussion
- Lab 1 check-in
- Building blocks of reproducible code

Anything to discuss? Questions?

What makes our code "reproducible"?

Thoughts on today's reading ("A Five Star Guide...")???

We've made some big leaps, but we haven't established a formalized vocabulary just yet

Let's start with indexing our vectors

```
library(tidyverse)

### same data: daily rainfall in inches
rainfall <- c(0.0, 2.1, 2.5, .1, 0.0, 0.0, 6.8, 3.1, 2.2)
```

How do we get the first element?

Indexing

```
# find the first element

rainfall[1]
```

Let's assume a "big" storm is one with 3" or more in a day... how do we check?

A simple logical test

```
# a VERY big storm is one 3" or greater, so let's check

rainfall[1] >= 3
```

what outputs are we limited to?

A simple if-else block

```
# Let's make an if-else block
if(rainfall[1] >= 3){
  print("big storm")
} else{
  print("little storm")
}
```

limitations of this method?

how does it generalize across the dataset?

Let's first wrap the if-else block in a function and replace the variables

```
f.storm.test <- function(rainfallAmount){
  if(rainfallAmount >= 3){
    print("big storm")
  } else{
    print("little storm")
  }
}
```

what does this code do?

Once you write a function, how do you use it?

A function with a loop

```
f.storm.test <- function(rainfallAmount){
  if(rainfallAmount >= 3){
    print("big storm")
  } else{
    print("little storm")
  }
}

for(i in rainfall){
  f.storm.test(i)
}
```

A note on loops

- R loops are inefficient
- Instead of appending the output of a loop to a dataset...
- they make a complete new copy
- So use them in small cases, not large ones

the `purrr::map` function is powerful, but confusing at times

the "tidy way"

```
rainfall %>% purrr::map(., f.storm.test)
```

How did your output differ?

There's also the good 'ol vectorized way of doing things

```
rainfall >= 3
```

if-else blocks vs. ifelse

```
if(sometest){
  do something in here
}else{
  do something else here
}
```

vs

```
ifelse(sometest, do something, do something else)
```

...functionally equivalent, really up to you

Finding the greatest rainfall day

Ideas?

A simple way

```
max(rainfall)
```

But which day is that?

The `which` command can be useful

Let's breakdown a weird syntax first...

```
which(rainfall == max(rainfall))
```

What happened?

A tidy way of working with data.frames

```
mydf <- read_csv("./data/ne_counties.csv")
glimpse(mydf)
```

Find the maximum median housing value

```r
max(mydf$MedValHous)
```

Which works the same way, but not super useful

```
which(mydf$MedValHous == max(mydf$MedValHous))
```

what's the output - and what does it mean?

Let's make it a bit more confusing

Break it down step-by-step

```
which(mydf$MedValHous == max(mydf$MedValHous)) %>% mydf[.,]
```

How exactly does this work?

Lastly, a "tidy" way

```
mydf %>% dplyr::slice_max(MedValHous)
```

A contrived question

for each county, calculate how much its median housing value is LESS than the max value in the dataset

Thoughts?

Easier than you'd think

Break it down

```
newdf <- mydf %>% mutate(deviation = MedValHous - max(MedValHous))
```

How would we find the county with the most negative deviation?

Like this!

```
newdf %>% dplyr::slice_min(deviation)
```

plot it

```
newdf %>% ggplot(., aes(x = deviation)) +
  geom_histogram() +
  theme_minimal()
```

Let's make it look a bit nicer

```
newdf %>% ggplot(., aes(x = deviation)) +
  geom_histogram(fill = "dark green") +
  theme_classic() +
  labs(title = "Deviations from maximum NE housing value",
       subtitle = "County scale",
       x = "Deviation",
       y = "Count")
```

...or just go a bit crazy

```
newdf %>% ggplot(., aes(x = deviation, y = after_stat(density))) +
  geom_histogram(fill = "dark green") +
  geom_vline(xintercept = mean(newdf$deviation), color = "red", linewidth = 2) +
  geom_density(color = "black", linewidth = 1) +
  theme_classic() +
  labs(title = "Deviations from maximum NE housing value",
       subtitle = "County scale",
       x = "Deviation",
       y = "Density")
```

What all has changed?

Review and next class

- Any questions?
- This week's readings/tasks:
  - Chapter 4 in textbook
  - Practice, practice, practice
  - Keep working on Lab 1